Applications of Derivatives

File 3 of 3 · Chapter-wise Practice Paper with Full Solutions

A chapter-wise paper in the Maharashtra HSC format (MCQ, VSA, 2-, 3- and 4-mark sections with internal choice). This is not a full official paper — it is 52 marks on Chapter 2 only. 18 items are verified board PYQs (Mar 2024 – Jun 2026, from your project folder), 6 are textbook questions and 5 are new board-style questions; each is labelled. Attempt the paper in 1 hr 45 min, then check the solutions. Mark splits shown are typical method-mark patterns, not an official scheme (none was available in the project files).

Prepared for Ammaar Khatri • Source: Balbharati textbook (2020, reprint 2022), pp. 65–94; board papers Mar 2024 – Jun 2026

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1 Instructions

- **Time:** 1 hour 45 minutes. **Maximum marks:** 52.
- **Section A:** Q.1 has six MCQs (2 marks each); Q.2 has four VSA questions (1 mark each). Attempt all.
- **Section B:** Q.3 – Q.10, 2 marks each. Attempt **any six**.
- **Section C:** Q.11 – Q.16, 3 marks each. Attempt **any four**.
- **Section D:** Q.17 – Q.21, 4 marks each. Attempt **any three**.
- Use of log table is allowed; calculator is not. Start each section on a new page.
- For MCQs only the first attempt counts, as in the board exam.

Source labels. **[PYQ month year Q.no.]** = verified from the board paper in your project folder. **[Textbook ...]** = Balbharati Part II, Chapter 2. **[New]** = written for this paper in board style; it is a practice question, not a prediction.

How to use this paper

Attempt everything in Part 1 under timed conditions with the solutions hidden. In the HTML file, solutions are collapsed; in the PDF they start in Part 2. Then mark yourself honestly using the mark splits, and read every examiner's note — including for questions you got right.

2 Part 1 — Question Paper

2.1 SECTION A

Q.1. Select and write the correct answer. [12]

(i) The slope of the tangent to the curve $x = \sin\theta$ and $y = \cos 2\theta$ at $\theta = \dfrac{\pi}{6}$ is _____. (2 · *Easy* · *parametric slope* · [*PYQ Mar 2024 Q1(v)*])

(a) $-2\sqrt{3}$ (b) $-\dfrac{2}{\sqrt{3}}$ (c) -2 (d) $-\dfrac{1}{2}$

(ii) The approximate value of $\tan(44°30')$, given that $1° = 0.0175^c$, is _____. (2 · *Medium* · *approximation* · [*PYQ Jul 2024 Q1(viii)*])

(a) 0.8952 (b) 0.9528 (c) 0.9285 (d) 0.9825

(iii) If the displacement of a particle at time t is $S = 2t^3 - 5t^2 + 4t - 3$, then its acceleration at $t = 1$ is _____. (2 · *Easy* · *acceleration* · [*PYQ Jul 2025 Q1(vi)*])

(a) 2 (b) 8 (c) 10 (d) 14

(iv) The approximate value of $f(x) = x^3 - 3x + 5$ at $x = 1.99$ is _____. (2 · *Easy* · *approximation* · [*PYQ Feb 2026 Q1(vi)*])

(a) 6.09 (b) 6.91 (c) 7.09 (d) 7.91

(v) If $f(x) = x^3 - 6x^2 + 9x + 18$, then $f(x)$ is strictly decreasing in _____. (2 · *Medium* · *increasing/decreasing* · [*Textbook Misc. 2 I(5)*])

(a) $(-\infty, 1)$ (b) $[3, \infty)$ (c) $(-\infty, 1] \cup [3, \infty)$ (d) $(1, 3)$

(vi) If $x = -1$ and $x = 2$ are the extreme points of $y = \alpha \log x + \beta x^2 + x$, then _____. (2 · *Hard* ·

stationary points · [*Textbook Misc. 2 I(6)*])

(a) $\alpha = -6,\ \beta = \frac{1}{2}$ (b) $\alpha = -6,\ \beta = -\frac{1}{2}$ (c) $\alpha = 2,\ \beta = -\frac{1}{2}$ (d) $\alpha = 2,\ \beta = \frac{1}{2}$

Q.2. Answer the following. [4]

(i) Write the condition for the function $f(x)$ to be strictly increasing for all $x \in \mathbb{R}$. (*1 · Easy* · [*PYQ Feb 2025 Q2(iv)*])

(ii) The displacement of a particle at time t is $S = 3t^2 + 4t + 1$. Find its velocity at time t. (*1 · Easy* · [*PYQ Jun 2026 Q2(iii)*])

(iii) Find the slope of the normal to $y = x^2$ at $(1, 1)$. (*1 · Easy* · [*New*])

(iv) If $f'(c) = 0$ and $f''(c) = 5$, what can you say about f at $x = c$? (*1 · Easy* · [*New*])

2.2 SECTION B — Attempt any SIX [12]

Q.3. Find the points on the curve $y = x^3 - 6x^2 + x + 3$ where the tangents are parallel to the line $y = x + 5$. (*2 · Easy · tangent parallel to a line* · [*PYQ Jul 2024 Q9*])

Q.4. Check whether the conditions of Rolle's theorem are satisfied by $f(x) = x^2 - 4x + 3$, $x \in [1, 3]$. (*2 · Easy · Rolle's theorem* · [*PYQ Jul 2025 Q11*])

Q.5. A stone is dropped into a quiet lake and waves in the form of circles are generated. The radius of the circular wave increases at the rate of 3 cm/sec. How fast is the area enclosed increasing when the radius is 8 cm? (*2 · Easy · rate measure* · [*PYQ Feb 2026 Q12*])

Q.6. Test whether the function $f(x) = x - \dfrac{1}{x}$, $x \in \mathbb{R}$, $x \neq 0$, is increasing or decreasing. (*2 · Easy · increasing/decreasing* · [*PYQ Jun 2026 Q9*])

Q.7. Divide the number 20 into two parts such that the sum of their squares is minimum. (*2 · Easy · optimisation* · [*PYQ Feb 2025 Q9*])

Q.8. The displacement of a particle at time t is given by $s = 2t^3 - 5t^2 + 4t - 3$. Find the time when the acceleration is 14 ft/sec². (*2 · Easy · motion* · [*PYQ Jul 2024 Q11*])

Q.9. Find the point on the curve $y = \sqrt{x - 3}$ where the tangent is perpendicular to the line $6x + 3y - 5 = 0$. (*2 · Medium · tangent perpendicular to a line* · [*Textbook Ex. 2.1 Q2*])

Q.10. Find the approximate value of $\sqrt{49.5}$. (*2 · Easy · approximation* · [*New*])

2.3 SECTION C — Attempt any FOUR [12]

Q.11. Find the equations of the tangent and normal to the curve $y = 2x^3 - x^2 + 2$ at the point $\left(\frac{1}{2}, 2\right)$. (*3 · Easy · tangent & normal* · [*PYQ Feb 2025 Q23*])

Q.12. Find the approximate value of $\tan^{-1}(1.002)$. [Given $\pi = 3.1416$] (*3 · Medium · approximation* · [*PYQ Mar 2024 Q22*])

Q.13. Water is being poured at the rate of 27 m³/sec into a cylindrical vessel of base radius 3 m. Find the rate at which the water level is rising. (*3 · Easy · rate measure* · [*PYQ Jul 2025 Q21*])

Q.14. Verify Lagrange's mean value theorem for the function $f(x) = \log x$ on $[1, e]$. (*3 · Medium · LMVT* · [*PYQ Feb 2026 Q22*])

Q.15. A wire of length 36 cm is bent to form a rectangle. Find its dimensions if the area of the rectangle is maximum. (*3 · Easy · optimisation* · [*PYQ Jun 2026 Q22*])

Q.16. Find the equations of the tangents to the curve $y = x^2 - 2x + 7$ which are (a) parallel to the line $2x - y + 9 = 0$, (b) perpendicular to the line $5y - 15x = 13$. (*3 · Hard · tangents with conditions* · [*New*])

2.4 SECTION D — Attempt any THREE [12]

Q.17. A box with a square base is to have an open top. The surface area of the box is 147 sq. cm. What should be its dimensions in order that the volume is largest? *(4 · Medium · optimisation · [PYQ Mar 2024 Q32])*

Q.18. A window is in the form of a rectangle surmounted by a semicircle. If the perimeter is 30 m, find the dimensions so that the greatest possible amount of light may be admitted. *(4 · Hard · optimisation · [Textbook Misc. 2 II(15)])*

Q.19. If Rolle's theorem holds for the function $f(x) = x^3 + px^2 + qx + 5$, $x \in [1, 3]$ with $c = 2 + \frac{1}{\sqrt{3}}$, find the values of p and q. *(4 · Hard · Rolle's theorem · [Textbook Ex. 2.3 Q4])*

Q.20. Show that the height of the cylinder of maximum volume that can be inscribed in a sphere of radius R is $\frac{2R}{\sqrt{3}}$. Also find the maximum volume. *(4 · Hard · optimisation · [Textbook Misc. 2 II(20)])*

Q.21. For $f(x) = 2x^3 - 21x^2 + 36x - 20$, (a) find the intervals in which f is strictly increasing and strictly decreasing; (b) find the local maximum and local minimum values. *(4 · Medium · increasing/decreasing + maxima/minima · [New], built on Textbook Ex. 2.4 Q9(ii))*

3 Part 2 — Solutions with Examiner's Notes

Solutions are written in the format expected in the answer book. Bracketed numbers like **[1]** show a typical split of method marks.

3.1 SECTION A — Solutions

Q.1 (i) — Answer: (c) -2

$\frac{dx}{d\theta} = \cos\theta,\ \frac{dy}{d\theta} = -2\sin 2\theta.$
$\frac{dy}{dx} = \frac{-2\sin 2\theta}{\cos\theta} = \frac{-4\sin\theta\cos\theta}{\cos\theta} = -4\sin\theta.$
At $\theta = \frac{\pi}{6}$: $\frac{dy}{dx} = -4\left(\frac{1}{2}\right) = -2$. **Answer (c).**
Examiner's note: writing $\sin 2\theta = 2\sin\theta\cos\theta$ cancels $\cos\theta$ and saves time. Option (a) comes from forgetting to divide by $\cos\theta$.

Q.1 (ii) — Answer: (d) 0.9825

$f(x) = \tan x$, $f'(x) = \sec^2 x$; $a = 45°$, $h = -30' = -\frac{0.0175}{2} = -0.00875^c$.
$f(a) = 1$, $f'(a) = \sec^2\frac{\pi}{4} = 2$.
$\tan(44°30') \approx 1 + (-0.00875)(2) = 0.9825$. **Answer (d).**
Examiner's note: h is negative because $44°30' < 45°$. Option (b) is a digit-swap distractor.

Q.1 (iii) — Answer: (a) 2

$v = \frac{dS}{dt} = 6t^2 - 10t + 4,\ a = \frac{dv}{dt} = 12t - 10$. At $t = 1$: $a = 2$. **Answer (a).**

Q.1 (iv) — Answer: (b) 6.91

$a = 2, h = -0.01$. $f(2) = 8 - 6 + 5 = 7$; $f'(x) = 3x^2 - 3$, $f'(2) = 9$.
$f(1.99) \approx 7 + (-0.01)(9) = 6.91$. **Answer (b).**

Q.1 (v) — Answer: (d) $(1, 3)$

$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$. $f'(x) < 0$ when $(x-1)$ and $(x-3)$ have opposite signs, i.e. $1 < x < 3$. **Answer (d).**
Examiner's note: (c) is where f is *increasing*.

Q.1 (vi) — Answer: (c) $\alpha = 2, \beta = -\frac{1}{2}$

$\frac{dy}{dx} = \frac{\alpha}{x} + 2\beta x + 1$. Extreme points $\Rightarrow \frac{dy}{dx} = 0$ at $x = -1$ and $x = 2$:
$-\alpha - 2\beta + 1 = 0$ and $\frac{\alpha}{2} + 4\beta + 1 = 0$.
From the first, $\alpha = 1 - 2\beta$. Substituting: $\frac{1-2\beta}{2} + 4\beta + 1 = 0 \Rightarrow 3\beta = -\frac{3}{2} \Rightarrow \beta = -\frac{1}{2}$, $\alpha = 2$.
Answer (c).

Examiner's note: this is a textbook MCQ; strictly, $\log x$ is undefined at $x = -1$, so the question is formal. In the exam, just do the algebra.

Q.2 — VSA answers

(i) $f'(x) > 0$ for all $x \in \mathbb{R}$.
(ii) $v = \dfrac{dS}{dt} = 6t + 4$.
(iii) $\dfrac{dy}{dx} = 2x = 2$ at $(1,1)$; slope of normal $= -\dfrac{1}{2}$.
(iv) f has a local minimum at $x = c$ (second derivative test: $f'(c) = 0$, $f''(c) > 0$).

3.2 SECTION B — Solutions

Q.3 — Points $(0,3)$ and $(4,-25)$

$y = x^3 - 6x^2 + x + 3 \Rightarrow \dfrac{dy}{dx} = 3x^2 - 12x + 1$. [½]
Slope of $y = x + 5$ is 1. Tangent parallel ⇒ slopes equal: [½]
$3x^2 - 12x + 1 = 1 \Rightarrow 3x(x-4) = 0 \Rightarrow x = 0$ or $x = 4$.
When $x = 0$, $y = 3$; when $x = 4$, $y = 64 - 96 + 4 + 3 = -25$. **[1]**
∴ Required points are $(0,3)$ and $(4,-25)$.
Examiner's note: students often stop at $x = 0, 4$. The question asks for **points** — both coordinates.

Q.4 — Conditions satisfied

$f(x) = x^2 - 4x + 3$ is a polynomial, hence continuous on $[1,3]$ and differentiable on $(1,3)$. **[1]**
$f(1) = 1 - 4 + 3 = 0$ and $f(3) = 9 - 12 + 3 = 0$, so $f(1) = f(3)$. **[½]**
∴ All the conditions of Rolle's theorem are satisfied. **[½]**
Examiner's note: the question only asks to *check* conditions; finding $c = 2$ is optional. Do not skip the reason ("polynomial").

Q.5 — 48π cm²/sec

Let R be the radius and A the area of the circular wave at time t. $A = \pi R^2$.
Differentiating w.r.t. t: $\dfrac{dA}{dt} = 2\pi R \dfrac{dR}{dt}$. **[1]**
Given $\dfrac{dR}{dt} = 3$ cm/sec. When $R = 8$ cm: $\dfrac{dA}{dt} = 2\pi(8)(3) = 48\pi$. **[1]**
∴ The area is increasing at the rate of 48π cm²/sec.

Q.6 — Strictly increasing

$f(x) = x - \dfrac{1}{x} \Rightarrow f'(x) = 1 + \dfrac{1}{x^2}$. **[1]**
For all $x \neq 0$, $x^2 > 0 \Rightarrow \dfrac{1}{x^2} > 0 \Rightarrow f'(x) > 0$. **[½]**
∴ f is an increasing function (strictly increasing on $(-\infty, 0)$ and on $(0, \infty)$). **[½]**
Examiner's note: the textbook answer is simply "increasing". Naming the two intervals is more precise: f is undefined at 0, and it is **not** increasing across 0 — for example $-0.5 < 1$ but $f(-0.5) = 1.5 > f(1) = 0$.

Q.7 — 10 and 10

Let the parts be x and $20-x$. Sum of squares $S = x^2 + (20-x)^2$.
$\frac{dS}{dx} = 2x - 2(20-x) = 4x - 40$. [½]
For extreme value $\frac{dS}{dx} = 0 \Rightarrow x = 10$. [½]
$\frac{d^2S}{dx^2} = 4 > 0$, so S is minimum at $x = 10$. [½]
∴ The two parts are 10 and 10. [½]

Q.8 — $t = 2$ sec

$s = 2t^3 - 5t^2 + 4t - 3$. Velocity $v = \frac{ds}{dt} = 6t^2 - 10t + 4$. [½]
Acceleration $a = \frac{dv}{dt} = 12t - 10$. [½]
Given $a = 14$: $12t - 10 = 14 \Rightarrow t = 2$. [**1**]
∴ The acceleration is 14 ft/sec² after 2 seconds.

Q.9 — Point $(4,1)$

$y = \sqrt{x-3} \Rightarrow \frac{dy}{dx} = \frac{1}{2\sqrt{x-3}}$. [½]
Slope of $6x + 3y - 5 = 0$ is $-\frac{6}{3} = -2$. Tangent perpendicular ⇒ its slope $= \frac{1}{2}$. [½]
$\frac{1}{2\sqrt{x-3}} = \frac{1}{2} \Rightarrow \sqrt{x-3} = 1 \Rightarrow x = 4, y = \sqrt{1} = 1$. [**1**]
∴ Required point is $(4,1)$.

Q.10 — $\sqrt{49.5} \approx 7.0357$

Let $f(x) = \sqrt{x}$, $f'(x) = \frac{1}{2\sqrt{x}}$. Take $a = 49$, $h = 0.5$. [½]
$f(49) = 7$, $f'(49) = \frac{1}{14}$. [½]
$f(a+h) \approx f(a) + hf'(a) \Rightarrow \sqrt{49.5} \approx 7 + \frac{0.5}{14} = 7 + 0.0357 = 7.0357$. [**1**]
Examiner's note: $0.5/14 = 0.035714\ldots$; four decimal places is enough. (True value $7.03562\ldots$)

3.3 SECTION C — Solutions

Q.11 — Tangent $2x - 4y + 7 = 0$; Normal $2x + y - 3 = 0$

$y = 2x^3 - x^2 + 2 \Rightarrow \frac{dy}{dx} = 6x^2 - 2x$. [½]
Slope of tangent at $\left(\frac{1}{2}, 2\right)$: $m = 6\left(\frac{1}{4}\right) - 2\left(\frac{1}{2}\right) = \frac{3}{2} - 1 = \frac{1}{2}$. [½]
Slope of normal $m' = -\frac{1}{m} = -2$. [½]
Equation of tangent: $y - 2 = \frac{1}{2}\left(x - \frac{1}{2}\right) \Rightarrow 4y - 8 = 2x - 1 \Rightarrow 2x - 4y + 7 = 0$. [**1**]
Equation of normal: $y - 2 = -2\left(x - \frac{1}{2}\right) \Rightarrow y - 2 = -2x + 1 \Rightarrow 2x + y - 3 = 0$. [½]

Q.12 — $\tan^{-1}(1.002) \approx 0.7864$

Let $f(x) = \tan^{-1} x \Rightarrow f'(x) = \dfrac{1}{1+x^2}$. [½]
Take $a = 1, h = 0.002$. [½]
$f(1) = \tan^{-1} 1 = \dfrac{\pi}{4} = \dfrac{3.1416}{4} = 0.7854;\ f'(1) = \dfrac{1}{1+1} = 0.5$. **[1]**
$f(a+h) \approx f(a) + hf'(a)$:
$\tan^{-1}(1.002) \approx 0.7854 + (0.002)(0.5) = 0.7854 + 0.001 = 0.7864$. **[1]**

Q.13 — $\dfrac{3}{\pi}$ m/sec

Let H be the height of water and V its volume at time t; base radius $R = 3$ m (constant).
$V = \pi R^2 H = 9\pi H$. **[1]**
Differentiating w.r.t. t: $\dfrac{dV}{dt} = 9\pi \dfrac{dH}{dt}$. **[1]**
Given $\dfrac{dV}{dt} = 27$: $27 = 9\pi \dfrac{dH}{dt} \Rightarrow \dfrac{dH}{dt} = \dfrac{3}{\pi}$. **[1]**
∴ The water level is rising at $\dfrac{3}{\pi}$ m/sec.
Examiner's note: substituting $R = 3$ before differentiating is correct here because the radius never changes. Contrast with $R = 8$ in Q.5, which must be substituted only after differentiating.

Q.14 — LMVT verified, $c = e - 1$

$f(x) = \log x$ is continuous on $[1, e]$ and differentiable on $(1, e)$, since $\log x$ is continuous and differentiable for all $x > 0$. So LMVT is applicable. **[1]**
$f(1) = \log 1 = 0,\ f(e) = \log e = 1,\ f'(x) = \dfrac{1}{x}$. [½]
By LMVT, $\exists\, c \in (1, e)$ such that $f'(c) = \dfrac{f(e) - f(1)}{e - 1}$:
$\dfrac{1}{c} = \dfrac{1-0}{e-1} \Rightarrow c = e - 1$. **[1]**
Since $2 < e < 3$, we have $1 < e - 1 < 2 < e$, so $c = e - 1 \in (1, e)$. Hence LMVT is verified. [½]

Q.15 — 9 cm × 9 cm (a square)

Let the length be x cm and breadth y cm. Perimeter $2(x + y) = 36 \Rightarrow y = 18 - x$. [½]
Area $A = xy = x(18 - x) = 18x - x^2$. [½]
$\dfrac{dA}{dx} = 18 - 2x$; for extreme value $18 - 2x = 0 \Rightarrow x = 9$. **[1]**
$\dfrac{d^2A}{dx^2} = -2 < 0$, so A is maximum at $x = 9$. [½]
Then $y = 18 - 9 = 9$. ∴ The rectangle is a square of side 9 cm. [½]

Q.16 — (a) $2x - y + 3 = 0$; (b) $12x + 36y - 227 = 0$

$y = x^2 - 2x + 7 \Rightarrow \dfrac{dy}{dx} = 2x - 2$. [½]
(a) Slope of $2x - y + 9 = 0$ is 2. Set $2x - 2 = 2 \Rightarrow x = 2, y = 4 - 4 + 7 = 7$. [½]
Tangent: $y - 7 = 2(x - 2) \Rightarrow 2x - y + 3 = 0$. [½]
(b) $5y - 15x = 13 \Rightarrow y = 3x + \frac{13}{5}$, slope 3. Perpendicular slope $= -\frac{1}{3}$. [½]
$2x - 2 = -\frac{1}{3} \Rightarrow x = \frac{5}{6};\ y = \frac{25}{36} - \frac{10}{6} + 7 = \frac{25-60+252}{36} = \frac{217}{36}$. [½]

Tangent: $y - \frac{217}{36} = -\frac{1}{3}\left(x - \frac{5}{6}\right)$. Multiply by 36: $36y - 217 = -12x + 10 \Rightarrow 12x + 36y - 227 = 0$. [½]
Examiner's note: the arithmetic in (b) is where marks go. Find y from the **curve**, not from the given line.

3.4 SECTION D — Solutions

Q.17 — Base 7 cm × 7 cm, height 3.5 cm

Let the side of the square base be x cm and the height h cm.
Surface area (open top) $= x^2 + 4xh = 147 \Rightarrow h = \dfrac{147 - x^2}{4x}$. ... (I) **[1]**
Volume $V = x^2h = x^2 \cdot \dfrac{147 - x^2}{4x} = \dfrac{147x - x^3}{4}$. [½]
$\dfrac{dV}{dx} = \dfrac{147 - 3x^2}{4}$. For maximum volume $\dfrac{dV}{dx} = 0 \Rightarrow x^2 = 49 \Rightarrow x = 7$ $(x \neq -7)$. **[1]**
$\dfrac{d^2V}{dx^2} = -\dfrac{6x}{4} = -\dfrac{3x}{2}$; at $x = 7$: $-\dfrac{21}{2} < 0$. $\therefore V$ is maximum at $x = 7$. [½]
From (I): $h = \dfrac{147 - 49}{28} = \dfrac{98}{28} = 3.5$. [½]
$\therefore$ For the largest volume the box has base 7 cm × 7 cm and height 3.5 cm (volume 171.5 cm³). [½]
Examiner's note: the commonest error is writing surface area as $2x^2 + 4xh$ — that is a **closed** box. Read "open top" twice. Check: optimal height is half the base side.

Q.18 — Radius $\frac{30}{\pi+4}$ m; rectangle $\frac{60}{\pi+4}$ m wide, $\frac{30}{\pi+4}$ m high

Let the radius of the semicircle be r m, so the width of the rectangle is $2r$; let the height of the rectangle be h m.
Perimeter $= h + 2r + h + \pi r = 30 \Rightarrow h = \dfrac{30 - 2r - \pi r}{2}$. ... (I) **[1]**
Light admitted $\propto$ area $A = 2rh + \dfrac{1}{2}\pi r^2 = r(30 - 2r - \pi r) + \dfrac{\pi r^2}{2} = 30r - 2r^2 - \dfrac{\pi r^2}{2}$. **[1]**
$\dfrac{dA}{dr} = 30 - 4r - \pi r$. For maximum, $\dfrac{dA}{dr} = 0 \Rightarrow r = \dfrac{30}{\pi + 4}$. **[1]**
$\dfrac{d^2A}{dr^2} = -(4 + \pi) < 0$, so A is maximum. [½]
From (I): $h = \dfrac{30 - r(2+\pi)}{2} = \dfrac{1}{2}\left[30 - \dfrac{30(2+\pi)}{\pi+4}\right] = \dfrac{15(\pi + 4 - 2 - \pi)}{\pi + 4} = \dfrac{30}{\pi+4}$.
$\therefore$ Width $= 2r = \dfrac{60}{\pi+4}$ m, height of rectangle $= \dfrac{30}{\pi+4}$ m, radius $= \dfrac{30}{\pi+4}$ m. [½]
Examiner's note: the perimeter of the window does **not** include the diameter (it is inside the window). Including it gives a wrong constraint $2h + 4r + \pi r = 30$.

Q.19 — $p = -6, q = 11$

$f(x) = x^3 + px^2 + qx + 5$. Since Rolle's theorem holds, $f(1) = f(3)$:
$1 + p + q + 5 = 27 + 9p + 3q + 5 \Rightarrow 8p + 2q = -26 \Rightarrow q = -13 - 4p$. ... (I) **[1]**
$f'(x) = 3x^2 + 2px + q$ and $f'(c) = 0$ with $c = 2 + \dfrac{1}{\sqrt{3}}$. [½]

$c^2 = 4 + \frac{4}{\sqrt{3}} + \frac{1}{3} = \frac{13}{3} + \frac{4}{\sqrt{3}} \Rightarrow 3c^2 = 13 + 4\sqrt{3};\ 2pc = 4p + \frac{2p}{\sqrt{3}}$. **[1]**
$f'(c) = 0:\ 13 + 4\sqrt{3} + 4p + \frac{2p}{\sqrt{3}} + q = 0.$
Using (I): $13 + 4\sqrt{3} + 4p + \frac{2p}{\sqrt{3}} - 13 - 4p = 0 \Rightarrow 4\sqrt{3} + \frac{2p}{\sqrt{3}} = 0 \Rightarrow p = -6$. **[1]**
From (I): $q = -13 + 24 = 11$. $\therefore p = -6, q = 11$. **[½]**
Check: $f'(x) = 3x^2 - 12x + 11 = 0 \Rightarrow x = \frac{12 \pm \sqrt{144 - 132}}{6} = 2 \pm \frac{1}{\sqrt{3}}$. ✓

Q.20 — $h = \frac{2R}{\sqrt{3}}$, $V_{\max} = \frac{4\pi R^3}{3\sqrt{3}}$

Let the cylinder have height h and base radius r. The centre of the sphere is the midpoint of the cylinder's axis, so by Pythagoras:
$r^2 + \left(\frac{h}{2}\right)^2 = R^2 \Rightarrow r^2 = R^2 - \frac{h^2}{4}$. **[1]**
$V = \pi r^2 h = \pi\left(R^2 h - \frac{h^3}{4}\right)$. **[½]**
$\frac{dV}{dh} = \pi\left(R^2 - \frac{3h^2}{4}\right) = 0 \Rightarrow h^2 = \frac{4R^2}{3} \Rightarrow h = \frac{2R}{\sqrt{3}}$. **[1]**
$\frac{d^2V}{dh^2} = -\frac{3\pi h}{2} < 0$, so V is maximum. **[½]**
$V_{\max} = \pi\left(R^2 - \frac{R^2}{3}\right)\frac{2R}{\sqrt{3}} = \pi \cdot \frac{2R^2}{3} \cdot \frac{2R}{\sqrt{3}} = \frac{4\pi R^3}{3\sqrt{3}}$ cubic units. **[1]**
Examiner's note: taking the half-height as the variable ($r^2 = R^2 - x^2$, $h = 2x$) is equally good and avoids fractions.

Q.21 — Increasing $x < 1$ or $x > 6$; decreasing $1 < x < 6$; max -3, min -128

$f(x) = 2x^3 - 21x^2 + 36x - 20 \Rightarrow f'(x) = 6x^2 - 42x + 36 = 6(x-1)(x-6)$. **[½]**
(a) $f'(x) = 0$ at $x = 1, 6$.

Interval	$x-1$	$x-6$	$f'(x)$	f
$x < 1$	$-$	$-$	$+$	strictly increasing
$1 < x < 6$	$+$	$-$	$-$	strictly decreasing
$x > 6$	$+$	$+$	$+$	strictly increasing

$\therefore f$ is strictly increasing on $(-\infty, 1) \cup (6, \infty)$ and strictly decreasing on $(1, 6)$. **[1½]**
(b) $f''(x) = 12x - 42$.
At $x = 1$: $f''(1) = -30 < 0 \Rightarrow$ local maximum; $f(1) = 2 - 21 + 36 - 20 = -3$. **[1]**
At $x = 6$: $f''(6) = 30 > 0 \Rightarrow$ local minimum; $f(6) = 432 - 756 + 216 - 20 = -128$. **[1]**
Examiner's note: (a) already tells you the answer to (b): increasing then decreasing at $x = 1$ means a maximum. Use this as a cross-check.

4 Answer key

Q	Answer	Q	Answer
1(i)	(c) -2	8	$t = 2$ s
1(ii)	(d) 0.9825	9	$(4, 1)$
1(iii)	(a) 2	10	7.0357
1(iv)	(b) 6.91	11	$2x - 4y + 7 = 0$; $2x + y - 3 = 0$
1(v)	(d) $(1, 3)$	12	0.7864
1(vi)	(c) $\alpha = 2,\ \beta = -\frac{1}{2}$	13	$\frac{3}{\pi}$ m/sec
2(i)	$f'(x) > 0$ for all x	14	$c = e - 1 \in (1, e)$
2(ii)	$v = 6t + 4$	15	9 cm × 9 cm
2(iii)	$-\frac{1}{2}$	16	$2x - y + 3 = 0$; $12x + 36y - 227 = 0$
2(iv)	local minimum at c	17	$7 \times 7 \times 3.5$ cm
3	$(0, 3)$, $(4, -25)$	18	$r = \frac{30}{\pi+4}$; width $\frac{60}{\pi+4}$, height $\frac{30}{\pi+4}$ m
4	satisfied ($c = 2$)	19	$p = -6, q = 11$
5	48π cm²/s	20	$h = \frac{2R}{\sqrt{3}}$, $V = \frac{4\pi R^3}{3\sqrt{3}}$
6	increasing	21	inc. $x < 1$ or $x > 6$; dec. $1 < x < 6$; max -3, min -128
7	10 and 10		

5 Source map: what is verified and what is new

Type	Questions
Verified board PYQ (from the papers in your project)	1(i) Mar 2024 · 1(ii) Jul 2024 · 1(iii) Jul 2025 · 1(iv) Feb 2026 · 2(i) Feb 2025 · 2(ii) Jun 2026 · 3 Jul 2024 · 4 Jul 2025 · 5 Feb 2026 · 6 Jun 2026 · 7 Feb 2025 · 8 Jul 2024 · 11 Feb 2025 · 12 Mar 2024 · 13 Jul 2025 · 14 Feb 2026 · 15 Jun 2026 · 17 Mar 2024
Textbook (Balbharati Ch. 2)	1(v) · 1(vi) · 9 · 18 · 19 · 20
New, board-style	2(iii) · 2(iv) · 10 · 16 · 21

Note: several PYQs are themselves textbook questions (e.g. Q.3 = textbook Ex. 2, p. 68; Q.8 = textbook Ex. 2, p. 71; Q.15 = Ex. 2.4 Q12). That overlap is exactly why the textbook is the best predictor of the paper — but no specific question is guaranteed to appear.

6 Formula checklist

Before the exam you should be able to write each of these from memory in under a minute:

- Tangent $y - y_1 = m(x - x_1)$; normal slope $-\frac{1}{m}$; slope of $ax + by + c = 0$ is $-\frac{a}{b}$.
- Parametric slope $\frac{dy/dt}{dx/dt}$ — and find the point.
- Related rates: formula → $\frac{d}{dt}$ → substitute; units.
- $v = s'$, $a = s''$, $j = s'''$.
- $f(a+h) \approx f(a) + hf'(a)$; $1^\circ = 0.0175^c$, $30' = 0.00875^c$.
- Rolle: three conditions ⇒ $f'(c) = 0$, $c \in (a, b)$. LMVT: two conditions ⇒ $f'(c) = \frac{f(b)-f(a)}{b-a}$.
- $f' > 0$ increasing, $f' < 0$ decreasing; sign table.
- $f'(c) = 0$: $f''(c) < 0$ max, $f''(c) > 0$ min, $f''(c) = 0$ test fails.
- Optimisation template: constraint → one variable → $f' = 0$ → f'' → answer with units.

7 Self-assessment guide

Score yourself out of 52 using the mark splits.

Score	What it means	What to do next
46 – 52	Board-ready for this chapter	Do the textbook Misc. Exercise 2 once more under time; move on
36 – 45	Solid, losing method marks	Re-read every examiner's note; redo the questions you lost marks on without looking
26 – 35	Concepts fine, execution weak	Use File 2's "Traps" list; redo Sections C and D after two days
below 26	Gaps in understanding	Go back to File 1 for the weak topics, then retake this paper in a week

Also check *where* marks were lost:

Lost marks mostly in ...	Likely cause	Fix
Section A / B	Speed, careless arithmetic	Re-derive slopes and rates slowly; verify points lie on the curve
Q.12, Q.10, Q.1(ii)	Approximation setup	Practise choosing a, h; radians for trig
Q.4, Q.14, Q.19	Rolle/LMVT write-up	Learn the 5-line answer template in File 1
Q.17 – Q.20	Optimisation setup	Draw the figure; write the constraint before anything else